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Case study

Modelling surface thermal damage to hot mill rolls

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Abstract

Surface thermal damage to hot mill work roll materials was investigated using twin-disc high-temperature laboratory test rig under conditions that are relevant to those of the early stands of a hot strip rolling mill. Two typical roll grades were tested: high speed tool steel and high-carbon high-chromium steel. Mill data were scaled down to suit the test sample dimensions and used to calculate the thermal cycle, contact times and contact pressure. A coupled thermomechanical model based on the finite element method was developed to predict the temperature distribution within the core and at the surface of the roll test samples. The predictions of the model showed good agreement with the temperatures measured using embedded thermocouples at several depths in the sub-surface of the roll test sample. After high temperature testing, the test samples were sectioned and prepared by conventional metallography for microscopic analysis. The thermally induced surface damage in the test samples were found to exhibit similar features to those observed in actual mill work rolls during service, especially the high-carbon-high-chromium steel, which showed a greater extent of surface thermal damage than the high speed tool steel test samples. Based on the results, it is concluded that the new test rig was proven capable of producing thermal damage in a consistent and reproducible manner and, thus, it can be regarded as a valuable off-line test resource to assist in further understanding the work roll surface deterioration phenomena.
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Keywords: Thermal damage; Hot rolling; Work roll; Laboratory simulation; Finite element modelling

1. Introduction

Work rolls are exposed to severe operating conditions where mechanical and thermal fatigue combine with tribological factors such as abrasion, adhesion and oxidation, all leading to surface wear. A work roll is subject to rapid changes of temperature on its surface layers due to the alternated contact with the hot strip and the cooling water in every rotation. As a consequence, temperature gradients are set upon the surface layers due to the relatively colder inner roll body. Cyclic thermal stresses and strains are imposed on the roll surface and, if the magnitude of the induced thermal stresses exceeds the hot yield strength of the material, then the roll working surface is likely to crack due to thermal shock and/or thermal fatigue. Thermal cracks are considered safe to remain at some depth threshold [1]; however, if these cracks are not removed by an appropriate roll dressing programme, they can lead to the propagation of larger cracks and subsequent spalling of roll surface fragments may occur [2]. Thermal fatigue and its associated roll surface failure mechanisms have a direct negative effect on the service life expectancy of the rolls and on the surface quality of the finished product, resulting in economic losses for the steel producers. Since the introduction of high speed steel (HSS) work rolls in the early 1990s to the finishing stands of hot strip mills, improved wear and thermal fatigue resistance have been observed compared to the traditional work roll materials but the progressive mechanisms of roll surface degradation are yet to be ascertained. Although a considerable amount of roll performance data can be obtained from full-scale trials, these experiments are technically difficult to perform in situ and are expensive, lengthy and inconvenient from the producer's point of view. In addition, rigorous levels of control and accurate measurements are necessary

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in order to achieve reliable and repeatable results. A more fundamental approach is laboratory simulation, which provides an attractive alternative in terms of relative cheapness, controllability of test conditions and quickness to acquire results compared with the full-scale industrial trials. This approach has been adopted by several investigators in the past 25 years. After a literature survey it was found that some 20 different purpose-built laboratory testing machines have been commissioned around the world aiming to study the wear of roll materials in both academic and industrial laboratories. Some of these machines are capable of performing tests at high temperature to simulate hot rolling conditions, amongst which there is a wide variety in designs and operating characteristics [3–13]. However, most of the research undertaken with these laboratory test machines has focussed on the sliding and abrasive wear of roll materials whereas thermal fatigue has remained relatively unstudied. It is in this context that a new hot roll wear and thermal fatigue simulation rig has been constructed, commissioned and used to model thermo-mechanical conditions during hot rolling of steel and to study the surface damage to hot mill rolls, focussing on their thermal fatigue failure. Several aspects of this investigation are presented and discussed in this paper.

2. Experimental equipment

The thermal fatigue experiments were performed using the new hot roll wear and thermal fatigue simulation rig shown in Fig. 1. This experimental facility is a high-temperature disc-on-disc type machine that allows testing in pure rolling and rolling/unidirectional sliding conditions. The top disc (hereafter "stock disc") simulates the hot steel being rolled while the bottom disc (hereafter "roll disc") mimics the roll. Each disc is independently driven via a frequency inverter to allow speed adjustment according to the desired amount of slippage during a test. A specially designed induction coil powered by a 15 kW high frequency solid state induction heating unit is employed to heat the stock disc working surface to a maximum achievable temperature of 1300 °C for 9 in. diameter stock discs after a heating time of approximately 6 min. A fibre optic two-colour pyrometer is employed to monitor the temperature of the running face of the stock disc. Once the desired hot rolling temperature has been achieved and the appropriate rotational speeds have been set, the two discs may be brought into contact using a hydraulic ram. The contact load may be displacement controlled or position controlled. A load cell monitors the applied load which is regulated to a fixed value via a loop to the servo-controller. Roll cooling is achieved by applying a flow of water onto the surface of the roll disc using a nozzle positioned at 90° after the contact. The complete disc assembly is enclosed in a sealable environmental chamber which can be filled with a mixture of nitrogen and air to allow a more accurate control of the oxide scale growth on the stock disc surface at high temperature to simulate descaling of the workpiece.

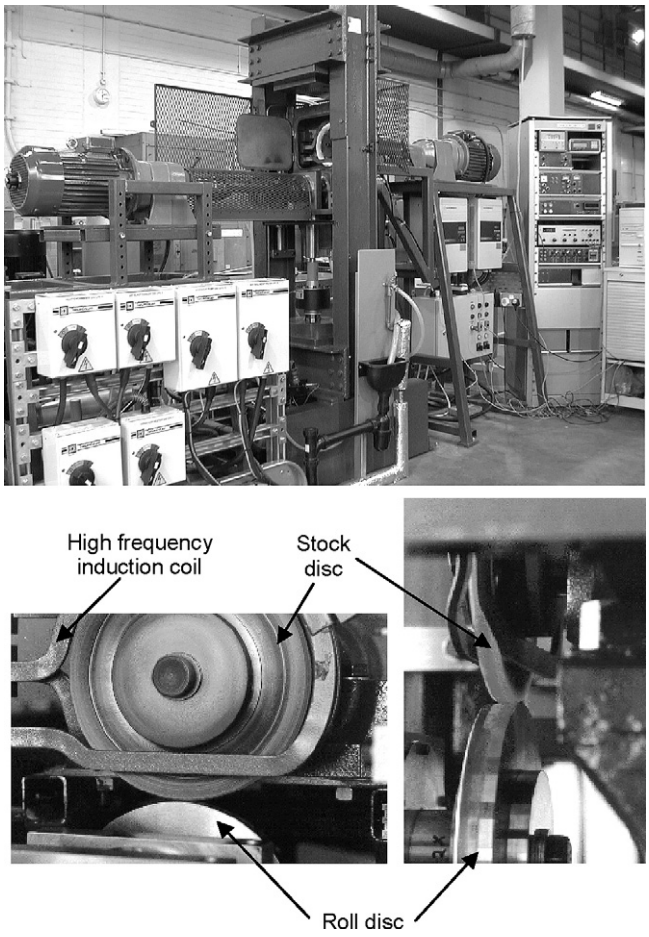


Fig. 1. The hot roll wear and thermal fatigue simulation rig.

3. Materials

The stock discs were manufactured from carbon steel En43A (BS970 Pt1). Roll discs were manufactured from two different alloys: high speed steel (HSS) and high carbon high chromium steel (HiCr). The chemical composition (wt%, balance Fe) of the test materials is presented in Table 1. The roll test pieces were manufactured as a complete roll sliced to the desired width, heat treated (hardened and tempered) and ground to the required specifications. Both roll and stock discs (shown in Fig. 2) are 228.6 mm (9 in.) in diameter but the stock discs are narrower than the roll discs (10 and 25 mm wide, respectively).

Table 1
Chemical composition of the stock and roll test materials used in this investigation (wt%, balance Fe)

	C	Si	Mn	P	S	Cr	V	Mo	Ni	W	Cu
Stock disc	0.49	0.22	0.85	0.023	0.02	–	–	–	–	–	–
HSS	1.5	0.66	0.54	0.018	0.017	7.77	4.9	2.08	0.52	0.49	0.06
HiCr	2.03	0.37	0.40	0.0.017	0.001	12.3	0.07	–	–	–	–

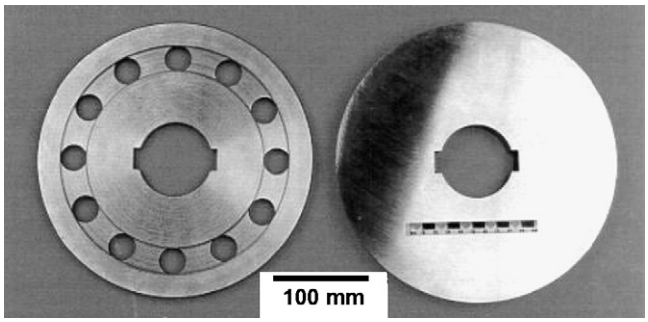


Fig. 2. Stock disc (left) and roll disc (right) test samples.

4. Calculation of the maximum contact pressure and stresses

Elastic pressures in the work rolls arising from their contact with the workpiece during hot rolling of steel typically lie within the range 100–350 MPa [7–9]. The maximum pressure p_o arising from the roll disc/stock discs contact configuration in this investigation was approximated to the case where two cylindrical bodies with parallel axes of radii R_1 and R_2, respectively are pressed together under a uniform specific constant force P per unit contact length l, assuming that the surfaces in contact are frictionless and their interface is free of tractions [14]. Thus:

$$p_o = \left(\frac{PE^*}{l\pi R}\right)^{1/2} \tag{1}$$

where the effective radius, R, is defined by Eq. (2) as:

$$R = \left(\frac{1}{R_1} + \frac{1}{R_2}\right)^{-1} \tag{2}$$

The effective Young's modulus, E^*, is given by Eq. (3):

$$E^* = \left(\frac{1-\nu_1{}^2}{E_1} + \frac{1-\nu_2{}^2}{E_2}\right)^{-1} \tag{3}$$

E_1 and E_2 are the Young's modulus of discs 1 and 2, respectively and ν_1 and ν_2 the Poisson's ratio of the two materials. For a pair of 229.6 mm diameter discs (i.e. $R_1 = R_2 = 114.8$ mm), the effective radius, R, is obtained by substituting into Eq. (2), thus obtaining $R = 57.4$ mm. For typical roll grades, Young's modulus, $E_1 = 225$ GPa, and Poisson's ratio, $\nu_1 = 0.29$ [15]. For carbon steel, $E_2 = 200$ GPa and $\nu_2 = 0.3$. By substituting these values into Eq. (3), the effective Young's modulus of the contact between the two test discs can be calculated as $E^* = 116$ GPa. The contact patch length l is equal to the width of the narrower of the contact discs (i.e. the stock disc, therefore $l = 10$ mm). By substituting these values into Eq. (1) and solving for P, it becomes possible to estimate the axial force required to produce maximum contact pressures ranging from 100 to 350 MPa, similar to hot rolling conditions. In this study, the simulation experiments were conducted under a constant axial contact load of 1.5 kN which results in a desired maximum contact pressure (p_o) of 310 MPa. The half-contact width b may be calculated using Eq. (4) while the distribution of the principal stress components σ_x, σ_y, σ_z and the principal shear stress τ_{yz} along the z-axis in the plane of symmetry of the loading curve may be calculated using Eqs. (5)–(8) [14]. Thus:

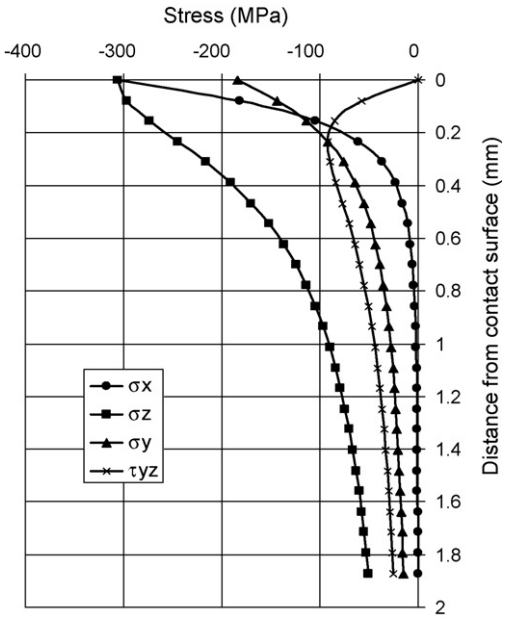


Fig. 3. Distribution of principal stress components and shear stress along the line of symmetry for $P = 1.5$ kN.

$$b = \left(\frac{4PR}{l\pi E^*}\right)^{1/2} \tag{4}$$

$$\sigma_x = -P_{max}\left[\left(1+\frac{2z^2}{b^2}\right)\left(1+\frac{z^2}{b^2}\right)^{-1/2} - 2\frac{z}{b}\right] \tag{5}$$

$$\sigma_z = -P_{max}\left(1+\frac{z^2}{b^2}\right)^{-1/2} \tag{6}$$

$$\sigma_y = \nu(\sigma_x + \sigma_z) \tag{7}$$

$$\tau_{yz} = -P_{max}\left[\frac{z}{b} - \frac{z^2}{b^2}\left(1+\frac{z^2}{b^2}\right)^{-1/2}\right] \tag{8}$$

The magnitude and position (along the z-axis) of the maximum principal shear stress $(\tau_{yz})_{max}$ may be calculated as [14]:

$$(\tau_{yz})_{max} = 0.304 p_o \tag{9}$$

$$z = 0.78b \tag{10}$$

Applying Eq. (4), with $P = 1.5$ kN, $R = 57.4$ mm, $l = 10$ mm and $E^* = 116$ GPa, the calculated half-contact width b is 0.31 mm. With Eqs. (8) and (9), for $p_o = 310$ MPa and $b = 0.31$ mm, the calculated maximum principal shear stress $(\tau_{yz})_{max}$ is 94.2 MPa at a distance $z = 0.242$ mm from the contact surface. The distribution of subsurface stresses in the contact zone along the line of symmetry calculated using Eqs. (5)–(8) are plotted in Fig. 3.

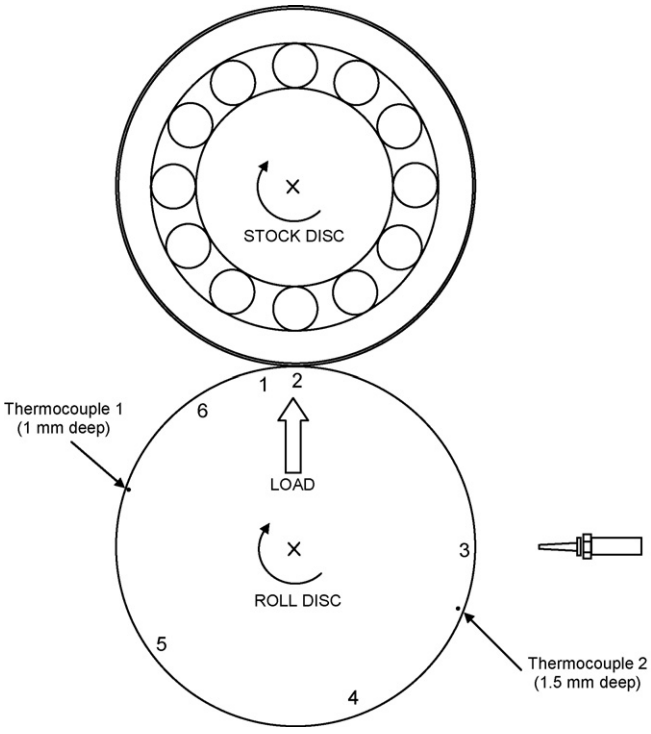


Fig. 4. Schematic diagram showing the position of the thermocouples for measuring the thermal cycle in the roll disc. Stock disc surface temperature was 1000 °C.

5. Thermal cycle measurement

Mill data taken from the first finishing stand of Corus hot strip rolling plant at Port Talbot were used to calculate the thermal cycle, contact times and contact pressure which are important factors when focussing on the thermal fatigue aspects of roll surface damage. These data were then scaled down to suit the test sample dimensions. Prior to conducting the full thermal fatigue testing programme, the thermal cycle at the sub-surface of the roll disc was measured using thermocouples connected to a PC-based data acquisition system. Two diametrically opposed thermocouples were embedded at 1 and 1.5 mm below the roll disc working surface as shown schematically in Fig. 4. The labels 1–6 in this figure indicate the locations where temperature changes are expected to occur due to the changes in the heat transfer conditions experienced by the roll disc surface at the various angular positions during one rotation. The temperature acquisition trial was performed at a constant contact load of 1.5 kN and at a constant rotational velocity of 6.4 rpm. To facilitate the insertion of the thermocouples, the environmental chamber was disabled, thus the temperature measurement trials were conducted in air until completion of 50 thermal cycles. The logged temperature obtained from this trial are shown in Figs. 5 and 6 which correspond to thermocouple depths of 1 and 1.5 mm, respectively from the roll disc running surface. Fig. 7 shows a closer view of thermal cycle development of the roll disc subsurface temperature recorded by both thermocouples placed at 1 and 1.5 mm deep from the roll disc running surface.

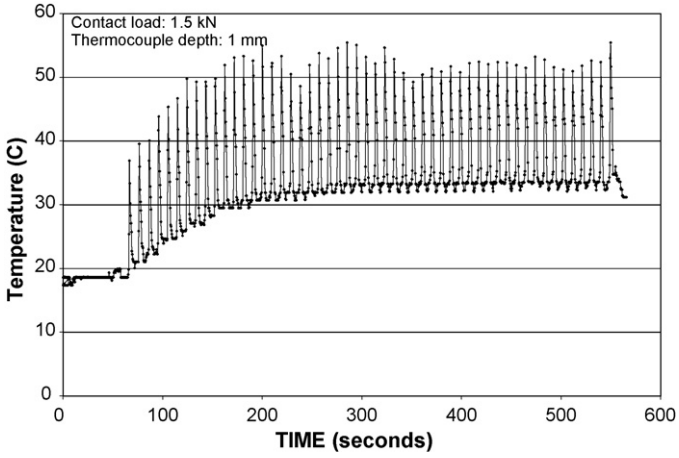


Fig. 5. Sub-surface thermal cycling of the roll disc test sample. Thermocouple depth: 1 mm, stock disc temperature: 1000 °C, contact load: 1.5 kN.

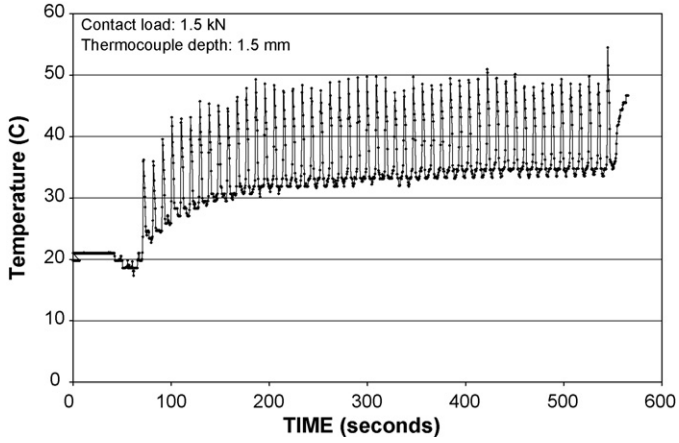


Fig. 6. Sub-surface thermal cycling of the roll disc test sample. Thermocouple depth: 1.5 mm, stock disc temperature: 1000 °C, contact load: 1.5 kN.

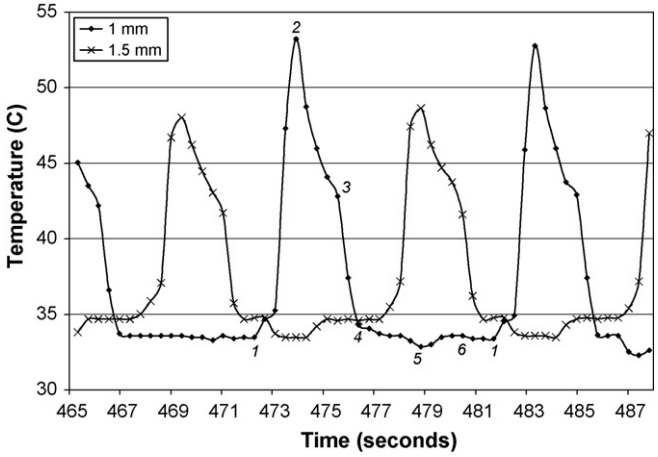


Fig. 7. Closer view of the thermal cycle development of the roll disc sample, contact load of 1.5 kN. Thermocouples were placed at a depth of 1 and 1.5 mm, respectively from the running surface. Labels 1–6 are referred as the angular locations shown in Fig. 4.

Table 2
Summary of testing conditions for the test specimens (all tests used roll disc water cooling and a nitrogen atmosphere)

Material	Load (kN)	Angular speed (rpm)	Stock disc temperature (°C)	Total number of rotations	Equivalent rolled distance (m)
High speed steel (HSS)	1–2	6.4[a]	950–1000	6280	4510
High carbon high chromium steel (HiCr)	1.5	6.4	1000	1824	1310

[a] The first 1800 rotations were performed at an angular speed of 20 rpm.

6. Testing programme

The thermal fatigue experiments were conducted in such a way that the thermal effects upon the roll disc specimen could be enhanced whilst effects of other nature (such as mechanical fatigue, sliding wear, abrasive wear, etc.) could be minimised. This provided reassurance that the damage observed in the roll disc surface after the experiments was mainly caused by thermal effects. In this context, the tests were conducted at a stock disc temperature of 1000 °C, which is representative of the strip temperature at the initial stands of a hot strip rolling mill. In order to eliminate traction forces at the roll/stock discs interface, no slippage was introduced at the contact, thus, minimising the likelihood of abrasive and/or sliding wear. Relatively long contact times allowed full development of the thermal cycle in the roll disc in each rotation. Thus, in order to achieve contact times that are relevant to those of the early stands of the hot strip mill, a rotational velocity of 6.4 rpm was chosen for this study. As described in the previous section, a constant contact load of 1.5 kN was used in the simulation experiments, resulting in a maximum contact pressure of 310 MPa, similar to the hot rolling scenario. The oxide scale grown on the stock disc surface at high temperature was kept as thin as possible in order to minimise the temperature loss across the oxide and across the stock/roll disc interface. To achieve this, the environmental control system was employed in all the thermal fatigue tests under a partial pressure of nitrogen. Roll disc water coolant was also used at a flow rate of 2.5 l/min in all the experiments. For the case of the HSS, a single sample disc was tested to a total of 6280 cycles, equivalent to a rolled distance of 4510 m. The high chromium high carbon steel was run to a total 1824 test cycles for an equivalent rolled distance of 1310 m. Upon tests completion, the discs were sectioned to an appropriate size and prepared using conventional metallography for microscopic analysis. A summary of the experimental programme employed in this study is given in Table 2.

7. Finite element modelling

The commercial package (finite element based) MSC.MARC/MENTAT was used for the construction and analysis of the modelling work carried out during this study. The approach used and some selected modelling outputs are given below.

(a) Geometric considerations

The different parts were created using external CAD (AutoCAD) and then imported into MSC.MARC/MENTAT where they were meshed and assembled to create a two-dimensional finite element model. Fig. 8(a) shows the final assembly.

(b) Material modelling

For the thermo-mechanical analysis, basic material properties for both high chromium and high speed alloys were used. These were: Young's modulus, Poisson's ratio, thermal expansion coefficient, conductivity, density and specific heat, all as a function of the temperature. Table 1 shows the mean composition of the work alloys.

(c) Mesh generation

To undertake the analysis both discs were subdivided into elements and the mesh density was refined along a surface region of 1 mm in depth (see Fig. 8b). In order to reduce computational time and due to the nature of the analysis a 2D model was created. Fig. 8 shows the finite element mesh used for the stock and roll discs. The final number of elements in the model approached 30,000.

(d) Initial and boundary conditions set-up.

The stock disc was considered to have an initial temperature of 1000 °C whilst the work roll disc was at room temperature. In order to describe appropriated thermal distributions over the model, different heat transfer coefficients were used for the various interfaces. For the stock-roll interface a heat transfer coefficient of 35 kW/m^{2}K was selected and to mimic the water cooling effect a convection coefficient of 5 kW/m^{2}K was chosen. In addition, the heat lost to the environment is taken into account using a value of 10 W/m^{2}K. Depending upon the model set-up both discs

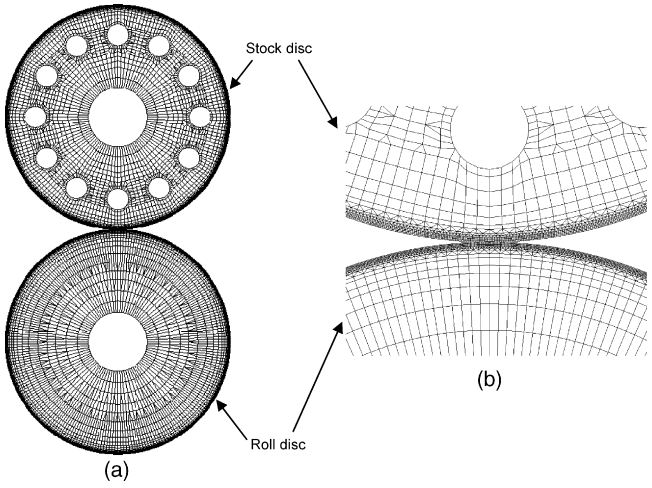


Fig. 8. (a) Final discs assembly; (b) close up of the contact zone of the finite element model.

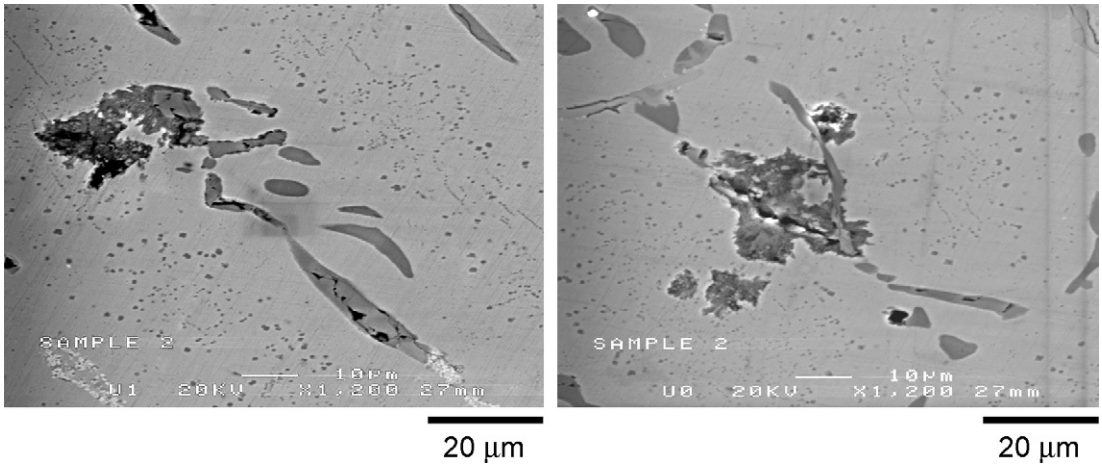


Fig. 9. SEM micrographs of the polished surface of the high speed steel roll disc after testing (6280 thermal cycles).

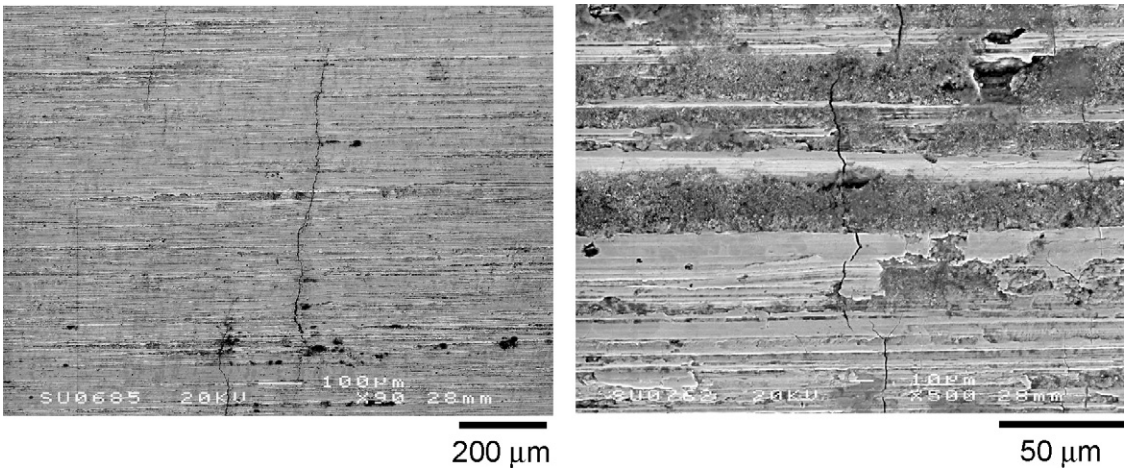


Fig. 10. SEM micrographs of the surface condition of the high-carbon high-chromium steel roll disc after testing (1824 thermal cycles).

came into contact at a particular load before rotation begins (see Table 2). Finally, a 2D shaft was created in order to rotate the roll disc; this was glued to the inner nodes of the disc and rotated at an angular speed of 6.4RPM.

8. Results and discussion

8.1. Experimental results

8.1.1. HSS

Fig. 9 shows the extent of thermal damage on the surface of the high speed steel roll disc after being tested for a total of 6280 thermal cycles. This sample was finely polished in order to visualize the cracking pattern on the surface of the roll disc sample. The extent of thermal damage imposed in this sample was far from severe, resulting only in a small number of minor micro-cracks within the carbides and along the carbide-matrix interface. Thus, cracks were only visible at very high SEM magnifications. Micro-cracks were not found within the matrix of the test sample. In line with the current observations, it has been reported in the literature that coarse carbides present on the surface of HSS thermal fatigue test specimens act as preferred crack initiation sites. This type of damage is commonly attributed to the appreciable difference in thermal expansion between the matrix and the carbides. Some evidence of carbide detachment was also observed in the test samples. The sub-surface region of

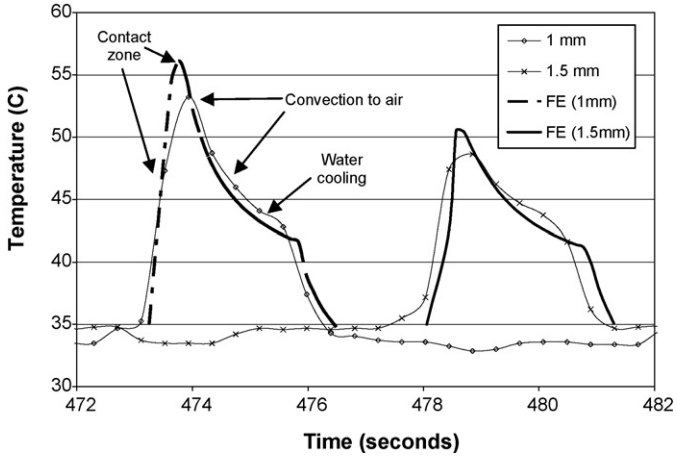


Fig. 11. Temperature comparison between experimental and modelling results after steady state conditions. Here, temperature readings at 1 and 1.5 mm underneath the surface are compared.

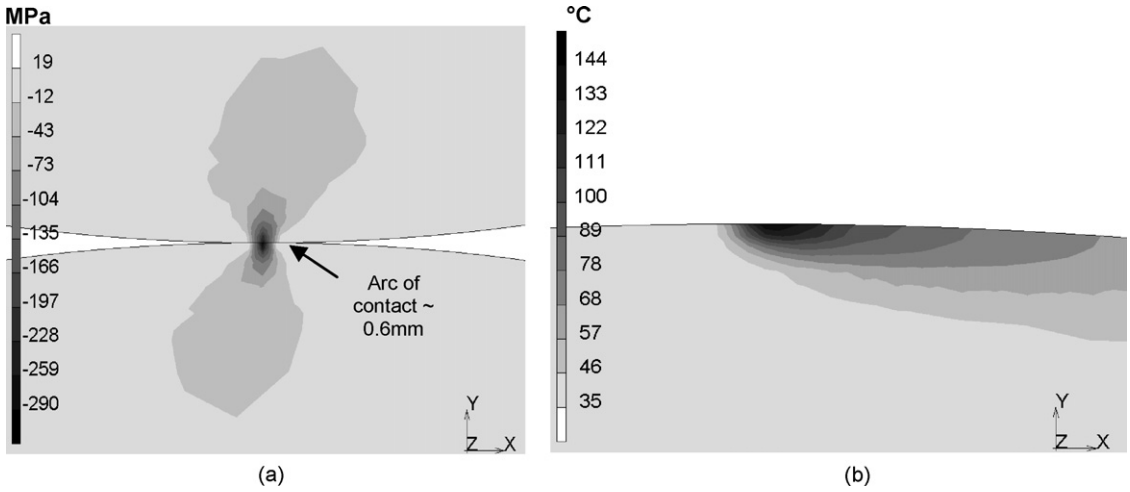


Fig. 12. (a) Normal stress distribution across the roll-stock interface predicted by the FE model. A peak compressive stress of about 290 MPa is observed. (b) Temperature distribution on the roll disc surface. A maximum temperature of about 144 °C is reached in each thermal cycle.

the sample was also analysed using SEM. Here, thermal micro-cracks were also limited to a small region of the order of size of the carbides. Micro-cracking was found only within the carbides as cracks did not propagate further into the matrix of the material.

8.1.2. HiCr

The surface condition of the high carbon high chromium (HiCr) test sample after a total of 1824 thermal cycles is shown in Fig. 10. In contrast with the HSS sample, the HiCr specimen exhibited thermal cracking to a larger extent as a fine network of thermal cracks were present after testing and these were readily visible even with the naked eye. The thermal cracks propagated perpendicularly to the rolling direction as the principal tensile stress at the surface (generally achieved during the cooling stage of the thermal cycle) acts along the circumferential direction of the roll disc test sample. Thermal crack coalescence and branching was repeatedly observed in this sample. Although cracks propagated to a large extent, the amount of damage is still small compared to the damage commonly observed in rolls. Cross-section sub-surface thermal cracks identified in the roll disc test samples initiated at the surface of the disc. Similarly to the HSS sample, coarse carbides acted as preferential sites for crack nucleation. Once nucleated, thermal cracks tend to propagate perpendicularly to the surface, in the radial direction along the matrix, following the carbide network. These experimental observations are in good agreement with the thermal crack morphologies developed by actual rolls in service reported in [16–20].

8.2. FEM results

Once a complete steady state run has been successful, different element quantities where selected as modelling output so that they could be analysed and interpreted. Fig. 11 shows a temperature comparison between experimental and modelling results; here the temperature predictions for the 1.5 kN load model are presented. The model predictions show a higher temperature peak than the experimental measurements. These differences are relatively small and are under the error tolerance of the thermocouples. It can also be observed from Fig. 11 that the main heat zones are also represented on the model predictions. Here, the heat gain by the work roll disc during contact gave a temperature rise of about 27 and 15 °C for 1 and 1.5 mm subsurface temperatures, respectively. The gradual heat los due to both air convection and water cooling was also observed. In general, the overall shape of the finite element predictions correlated well with the experimental readings. Fig. 12(a) shows the normal stress predictions for the contact zone. Here the peak compressive stress reaches 290 MPa. This stress correlates well with the predictions shown in Section 4. In addition, the deformation imposed by the load created and arc of contact of the order of 0.6 mm. The temperature distribution at the surface of the roll disc as predicted by the finite element model is presented in Fig. 12(b). Here, a maximum temperature of about 144 °C is reached in each thermal cycle. It becomes clear from this prediction that the magnitude of the maximum temperature involved in each cycle is much lower than in the actual hot rolling scenario. This is mainly attributed to the relatively low contact time and the small area of contact at the roll/stock discs interface during the experiments. However, the temperature fluctuation at the roll disc surface was sufficient to induce thermal fatigue cracking on the surface of the sample.

9. Conclusions

A new test rig has been developed to study surface thermal damage to roll materials for hot rolling of steel at a laboratory scale under conditions that are well controlled and monitored. Experiments were conducted in two typical roll material grades: HSS and HiCr. Both materials exhibited thermal cracking after testing to some extent. The tests performed on the HSS material did not produce appreciable surface degradation, as the only thermal damage observed was in the form of micro-cracks within the carbides or along the carbide–matrix interface. The thermal damage imposed on the HiCr test sample was more severe,

exhibiting large thermal cracks that were comparable to those encountered in actual rolls during service. The experimental set-up was modelled by a coupled thermo-mechanical finite element model, which is representative of the actual experimental trials and predicts both temperature and stress distributions with good agreement. The support of the finite element analysis is crucial in order to have a complete understanding of the phenomena at the roll stock interface. Based on the results, the new test rig was proven capable of producing thermal damage in a consistent and reproducible manner and, thus, it can be regarded as a valuable off-line test resource to assist in further understanding the work roll surface deterioration phenomena.

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